A new project, MyBlockchain in Truffle can be mapped as follows:

```
mkdir MyBlockchain
cd MyBlockchain
truffle MyBlockchain
```

In Truffle, the directory structure given below is followed to code the application:

- *contracts/:* Source code for the smart contracts
- *migrations/:* Migration system and handlers for the smart contracts
- *test/:* Tests and JavaScript code
- *truffle.js:* Configuration file for Truffle
- *MyBlockchain:* Additional folders and files required for coding the blockchain

A new file <*filename.sol*> is created in the *contracts/* directory for the base coding of smart contracts in the following format:

```
pragma solidity ^0.5.0;
contract Mysmartcontract {
}
Sending Values
function adopt(uint MyVar) public returns (uint) {
  require(MyVar >= 0 && MyVar <= 15);
  adopters[MyVar] = msg.sender;
  return MyVar;
}
```

The compilation of Truffle code is done as follows:

```
truffle compile
```

Embark

The Embark framework provides the tools and libraries for the development of decentralised apps so that blockchain based implementation can be executed. Embark can be used as an alternate to Truffle. To work with Embark, you need to integrate Node Version Manager (NVM) with multiple versions of NodeJS.

NVM can be installed as follows:

```
curl -o- https://raw.githubusercontent.com/creationix/nvm/
v0.33.11/install.sh | bash
nvm install --lts
nvm use --lts
```

Once NVM is installed, the toolkit of Embark can be associated as follows:

```
npm -g install embark
```

After installation, the initialisation instruction is executed for the new project, as follows:

```
embark new mysmartcontract
cd mysmartcontract
```

By using 'embark run', the dashboard of Embark is invoked.

The token generation can be further written in code as follows:

```
pragma solidity ^0.4.25;
import "openzeppelin-solidity/contracts/token/ERC20/ERC20.
sol";
contract CryptoToken is ERC20 {
    string public name = "CryptoToken";
    string public symbol = "MC";
    uint256 public decimals = 18;
        constructor() public {
    }
}
```

As in the above example, MC refers to the new currency symbol, which can be anything, depending on the requirements of the smart contract associated with the blockchain.

Scope for research and development

As blockchain based development is an emerging domain of research, there is a need to address different issues related to privacy and resource optimisation. In blockchain and other decentralised applications, the data is replicated to numerous devices and the issues of security and integrity do not arise. With the development and deployment of advanced algorithms, the performance of blockchain based implementations can be elevated further. END

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